initially thought. Einstein did not want to relent on what was for him the key issue: that there was an objective reality independent of whoever interacts with whatever. Bohr would not relent on the validity of the profoundly new way in which the real was conceptualized by the new theory. Ultimately, Einstein conceded that the theory was a giant step forward in our understanding of the world, but remained convinced that things could not be as strange as it proposed – that 'behind' it there must be a further, more reasonable explanation.

A century later we are at the same point. The equations of quantum mechanics and their consequences are used daily in widely varying fields: by physicists, engineers, chemists and biologists. They are extremely useful in all contemporary technology. Without quantum mechanics there would be no transistors. Yet they remain mysterious. For they do not describe what happens to a physical system, but only how a physical system affects another physical system.

What does this mean? That the essential reality of a system is indescribable? Does it mean that we only lack a piece of the puzzle? Or does it mean, as it seems to me, that we must accept the idea that reality is only interaction? Our knowledge grows, in real terms. It allows us to do new things that we had previously not even imagined. But that growth has opened up new questions. New mysteries. Those who use the equations of the theory in the laboratory carry on regardless, but in articles and conferences that have been increasingly numerous in recent years physicists and philosophers continue to search. What is quantum theory, a century after its birth? An extraordinary dive deep into the nature of reality? A blunder that works, by chance? Part of an incomplete puzzle? Or a clue to something profound regarding the structure of the world which we have not yet properly digested?

When Einstein died, his greatest rival Bohr found for him words of moving admiration. When a few years later Bohr in turn died, someone took a photograph of the blackboard in his study. There's a drawing on it. A drawing of the 'light-filled box' in Einstein's thought experiment. To the very last, the desire to challenge oneself and understand more. And to the very last: doubt.